

Informative censoring does not cancel in an unanchored comparison, and adjusting one side can make it worse

A simulation study answering catalog problem OUT-07

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Abstract

Background. When censoring depends on patients' risk, Kaplan-Meier estimates in each trial are biased. It is often argued that the biases cancel in an indirect comparison. Catalog problem OUT-07 asks when they do.

Methods. A shared frailty drove both event and censoring hazards, with separate dependence in the source (none or moderate) and the comparator (none, moderate or strong), censoring of 20% or 50%, in unanchored and anchored comparisons: 24 scenarios, 1000 replicates each. Estimand: RMST difference at 24 months (0.62 months unanchored). Methods: Kaplan-Meier, inverse probability of censoring weighting on the source side with a baseline proxy of risk, and a one-sided comparator sensitivity analysis.

Results. Unanchored, with differing mechanisms the bias ranged from -2.33 to 1.42 months, and the sign of the comparison was wrong in up to 0.973 of analyses. Equal mechanisms nearly cancelled (-0.080 to -0.030). Anchored comparisons were protected in every scenario (-0.202 to 0.219), because both arms of each trial shared its mechanism. Source-side weighting reduced source bias only partly and, where the mechanisms were equal, broke the cancellation (bias -0.080 to -0.515 at 50% censoring). The comparator sensitivity parameter that reproduced the truth lay outside the conventional range 0.5 to 2 in up to 0.703 of analyses.

Conclusion. Cancellation needs equal mechanisms, which an external or single-arm comparator seldom has. Anchoring protects; adjusting the source alone does not, and a one-sided sensitivity analysis must allow larger departures than usual.

1 The problem

Each Kaplan-Meier RMST (1) carries a censoring bias c ; an unanchored contrast carries $c_S - c_T$ and an anchored one the difference of within-trial differences, in which each trial's shared mechanism largely cancels. Inverse probability of censoring weighting (2) needs the censoring predictors, which exist for the individual-data source and not for an aggregate comparator.

2 Design

Registered protocol: `protocol.md`; ADEMP structure (3). Frailty $U \sim N(0, 1)$; event hazard $\lambda e^{0.7U}$ ($\lambda_A = 0.045$, $\lambda_B = 0.05$, $\lambda_C = 0.07$ per month); censoring hazard $\kappa e^{\alpha U}$, κ set for the censoring prevalence by 24 months; administrative end at 36; 300 per arm. The source records $W = 0.7U + \text{noise}$. The sensitivity analysis gives censored comparator patients a remaining hazard of δ times the arm's crude rate ($\delta = 1$ is independent censoring).

3 Results

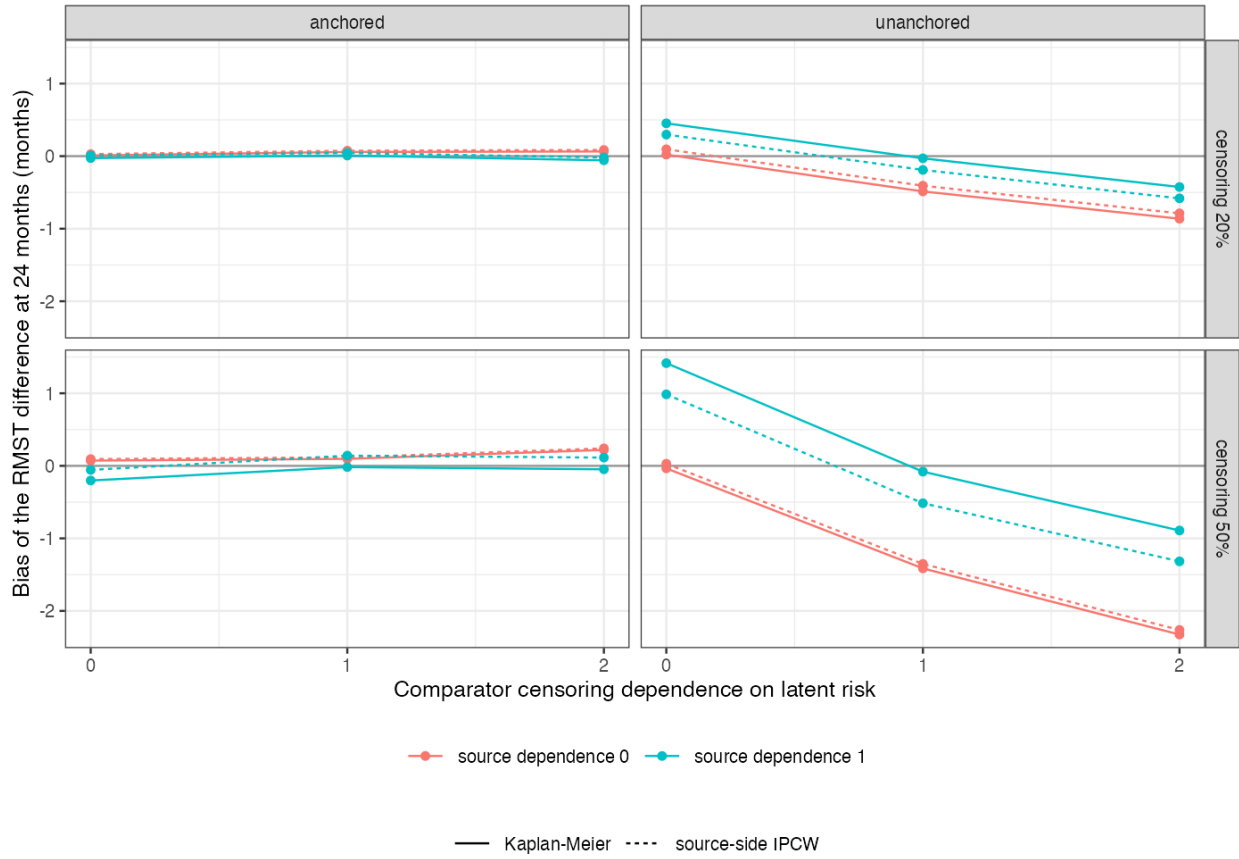


Figure 1: Bias of the RMST difference by comparator censoring dependence, for each level of source dependence.

Unanchored bias followed $c_S - c_T$ (Figure 1): positive when only the source censored its high-risk patients, negative when only the comparator did, and roughly twice as large at 50% censoring. With a true difference of 0.62 months, a comparator with strong dependence reversed the comparison in 0.973 of analyses at 50% censoring.

Weighting on a proxy correlated 0.7 with the latent risk removed about a third of the source bias. Applied only to the source, it turned a cancelled comparison into a biased one. In the sensitivity analysis the δ that reproduced the truth had a median of 2.43 with strong comparator dependence, so a range capped at 2 would often not reach it.

4 What this does not answer

No covariate shift between trials, so the population-adjustment layer is absent; time-constant censoring hazards; one sensitivity parameterization; a single baseline proxy stands in for the source’s time-varying censoring predictors. Peer review has not been done.

References

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